

Suicidal Sentiments Identifier

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Abstract

Social media and messaging platforms are at the center of understanding human behavior. Our aim is creating a machine learning model that detects suicidal tendencies in communication, typically social media posts and messages, which are in a text form. The model classifies the users through speech patterns and surrounding circumstances using a classification model; exploring methods such as Support Vector Machines, Naive Bayes, etc. The method ensures a powerful tool to save lives and intervene in the health-threatening behaviors of individuals.

1. Motivation

Suicidal behaviour is a serious global problem; it takes over 700,000 lives a year, placing it as one of the leading causes of death in the world. Suicide is the fourth most common cause of death among young people between the ages of 15 and 29 years. There is great awareness about mental health in today's society, but identifying people at risk has become very challenging. The motivation of this project lies in finding an effective and scalable solution for early identification of suicidal tendencies through machine learning, which can save many lives with timely intervention.

2. Related Work

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9565132/>: This paper uses data of tweets posted by individuals on twitter. It uses various DL methods and ML classifiers and conducts a binary classification to detect tweets that may have suicidal tendencies.

https://www.researchgate.net/publication/323065338_Online_suicide_prevention_through_optimised_text_classification:

This paper investigates a method to automatically classify suicide-related Dutch forum posts. They used Naive Bayes and Support Vector Machine for classification.

https://www.researchgate.net/publication/259948193_Predicting_Depression_via_Social_Media: This paper explores social media investigate Major Depressive Disorder among twitter users.

3. Timeline

1. **Week 4:** (2nd September - 6th September) - Find and prepare a dataset for analysis.

2. **Week 5:** (9th September - 13th September) - Explore dataset for any outliers and try to clean noise out if any.
3. **Week 6-7:** (17th September - 30th September) - Experiment with different classification models.
4. **Week 8-9:** (14th October - 24th October) - Address data imbalances.
5. **Week 10-11:** (4th November - 14th November) - Test various regression models and apply them.
6. **Week 12-13:** (18th November - 29th November) - Combine outcomes for final analysis and conclusion.

4. Individual Contributions

All members will contribute equally to the workload, exploring various machine learning techniques to detect suicidal tendencies using only text-based communication. We hope to study methods such as Support Vector Machines, Naive Bayes, and Logistic Regression and take advantage of the expertise within each member later down the course. We would make the decisions together on which methods to use and distribute the workload based on constraints in time, availability, and expertise. We have an aim to develop a robust model regarding early identification and intervention.

5. Final Outcome

We plan to use this project to predict whether a particular piece of text was written or posted by someone who may be experiencing suicidal thoughts. This will enable timely responses and support.

References

- [1] T. Aldhyani, S. N. Alsubari, A. S. Alshebami, H. Alkahtani "Detecting and Analyzing Suicidal Ideation on Social Media Using Deep Learning and Machine Learning Models", <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9565132/>
- [2] Nikhileswar Komati, Suicide and Depression Detection, Retrieved from [Kaggle](#)